SPE-229100-MS

# Mobile Solar Panels and Diesel Engine Hybrid Power Generating for Drilling Rigs

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.



## Abstract

This paper presents the design, deployment, and performance evaluation of a mobile solar-diesel hybrid energy system deployed by an onshore oil and gas Operator in the United Arab Emirates (UAE) and used to power a land drilling rig in an off-grid location in Abu Dhabi, UAE. Traditionally powered solely by diesel generators, this rig was the site of a 5-month pilot project testing the viability of a containerized mobile solar hybrid integrated system in high-demand upstream operations. The system was comprised of 4 × 75 kWp solar containers (each with integrated inverters), diesel generators matched to variable rig loads, and a custom-built hybrid energy management system (EMS). Operating as a plug-and-play "energy as a service" solution, the system ensured uninterrupted sustainable power for rig operations.

Over the pilot period (July-November 2024), solar power contributed 117 MWh (7% of total rig energy demand), directly reducing diesel consumption and associated $CO_2$ emissions. Though the solar share was modest due to the rig's high load profile, the hy brid solution delivered improved fuel efficiency, reduced acoustic impact, minimized generator maintenance, and effective solar share during non-drilling operations. Moreover, this pilot project served as a proof of concept for using mobile solar solutions to power high demand onshore drilling operations. The containerized design enabled rapid mobilization (under 6 hours) and remote diagnostics which significantly reduced onsite support requirements.

This project marks one of the region's first mobile solar-diesel hybrid integrations for powering an active rig and offers practical insights for scaling sustainable solutions in upstream operations. It provides valuable operational insights into the potential for hybrid power systems to support decarbonization targets in high-demand upstream environments. The successful operation over the 5-month period underscores the feasibility of scaling this solution to multiple rigs across the Operator's portfolio while adding battery storage capabilities to increase diesel savings. The project highlights the Operator's proactive approach toward sustainable energy practices, aligned with national and global net-zero targets for 2045.

## Introduction

In the broader context of global energy transition, hybrid renewable energy systems are emerging as a critical enabler for reducing dependency on fossil fuels in high-demand industrial applications. Oil and gas operations, particularly in off-grid environments, face a unique set of challenges that make continuous, reliable power generation both essential and costly. The integration of solar PV with diesel generation addresses these challenges by optimizing fuel usage, lowering operational costs, and contributing to emissions reduction targets. The United Arab Emirates, with its high solar irradiance and commitment to net-zero by 2045[1], provides an ideal setting for testing and scaling such solutions.

### Problem Statement

As the world transitions toward a more sustainable future, the oil and gas industry must take an active role in reducing emissions and achieving net-zero targets by 2045. To support these goals, companies are prioritizing the reduction of methane emissions, enhancing operational efficiency, adopting carbon capture and storage technologies, and integrating on-site renewable energy sources. A key challenge in this transition for oil and gas companies is providing reliable and sustainable power for off-grid operations—such as drilling rigs, artificial lift stations, base camps, and mobile camps.

The UAE Operator highlighted in this paper sets a strong example by implementing practical decarbonization strategies, especially in off-grid environments. Historically, their drilling rigs across the UAE have depended on large, inefficient diesel generators, which run counter to sustainability objectives and present issues such as poor reliability, high emissions, inefficiency, and elevated operational costs. These challenges also negatively affect health, safety, environmental, and governance (HSE&G) performance indicators. Remote upstream drilling operations continue to rely heavily on these diesel generators. Although effective, this approach results in high fuel consumption, substantial carbon dioxide ($CO_2$) emissions, and complex logistics—particularly for mobile drilling rigs that frequently move between remote sites. As the global energy sector advances toward decarbonization, integrating renewable energy solutions into off-grid drilling operations is becoming increasingly important.

In the UAE, where achieving net-zero emissions by 2045 is a national ambition, the task is made more difficult by extreme climate conditions and infrastructure limitations. In response, the Operator featured in this paper has taken a pioneering step by piloting a mobile solar-diesel hybrid system to power an active onshore drilling rig in Abu Dhabi, UAE. This marks both a regional first and a significant move away from traditional diesel-only generator setups.

### Objective

The objective of this project was to implement and evaluate the feasibility and performance of a mobile solar-diesel hybrid system in powering an active land drilling rig without compromising reliability. Key metrics included solar contribution, diesel fuel displacement, $CO_2$ reduction, ease of deployment, system reliability, and operational stability. The system was designed for modularity, mobility, and real-time remote operation, and was delivered as an integrated "energy as a service" solution that can be integrated with a battery storage capacity to further increase the decarbonization metrics.

## Technology: Mobile Solar-Diesel Hybrid System

### System Overview

The mobile solar hybrid system consists of four mobile solar containers (including inverters), a control panel (energy management system EMS and data loggers), three diesel generators (two prime and one standby, with capacities adjusted for drilling activity load variations), and a fuel tank with a level sensor. The energy management system (EMS), developed in-house by the system Provider, maximizes solar utilization, and ensures efficient operation of energy sources. The system features advanced remote monitoring capabilities, nearly eliminating the need for frequent physical intervention. The components and layout of the system will be detailed in the subsequent section.

The deployed system includes:

- 4 × 75 kWp mobile solar containers (total 300 kWp)
- Integrated solar inverters within each container (66 kVA per solar container)
- Diesel generators (prime and backup units) scaled to rig load
- A hybrid control panel and custom Energy Management System (EMS)
- Real-time remote monitoring platform, including gateways for load and solar metering
- Optional interface for future battery integration

The Energy Management System (EMS), developed by the Provider, manages the energy sources in real time. It prioritizes solar input while ensuring generator operation at optimal efficiency and microgrid stability under variable rig loads. This system is designed to be modular which allows for more solar capacity to be installed in addition to battery storage capabilities which helps improve the solar share and diesel consumption reduction.


**Figure 1—Mobile Solar Hybrid System Deployed at site in Abu Dhabi, UAE**

## Component Integration and Layout

Each solar container houses retractable PV arrays and associated power electronics. The hybrid controller synchronizes with the rig's load profile and manages transitions between energy sources. Diesel generators run in synergy with solar input, reducing fuel consumption without compromising power availability.

**Key integration features include:**

- Plug-and-play connection to rig Low Voltage (LV) panel
- Modular design for easy scaling or redeployment
- Rapid installation and refolding (under 6 hours)
- Weather-resistant design suited for desert conditions
- Adherence to Operator's standard operating procedures
- Minimal groundwork and civil work preparation
- No interference with site operations, rig up, rig down, and move times

A future-ready configuration allows for seamless integration of Battery Energy Storage Systems (BESS) to further enhance diesel displacement and load stability.

The integration and management of the solar-diesel hybrid system are crucial to its success. The hybrid manager, a key component, facilitates communication between generators and solar inverters for efficient power supply. The energy management system (EMS) communicates with various sensors (fuel, temperature, irradiance) and energy meters (load requirements, diesel generator energy, and renewable energy) through three local data buses. Live plant data is accessible via a remote monitoring system, allowing for proper monitoring and rapid response to onsite events. Data visualization, warehousing, and performance analysis are conducted using the Provider's in-house online platform. The Operator also has full access to project data through a client portal developed by the Provider.

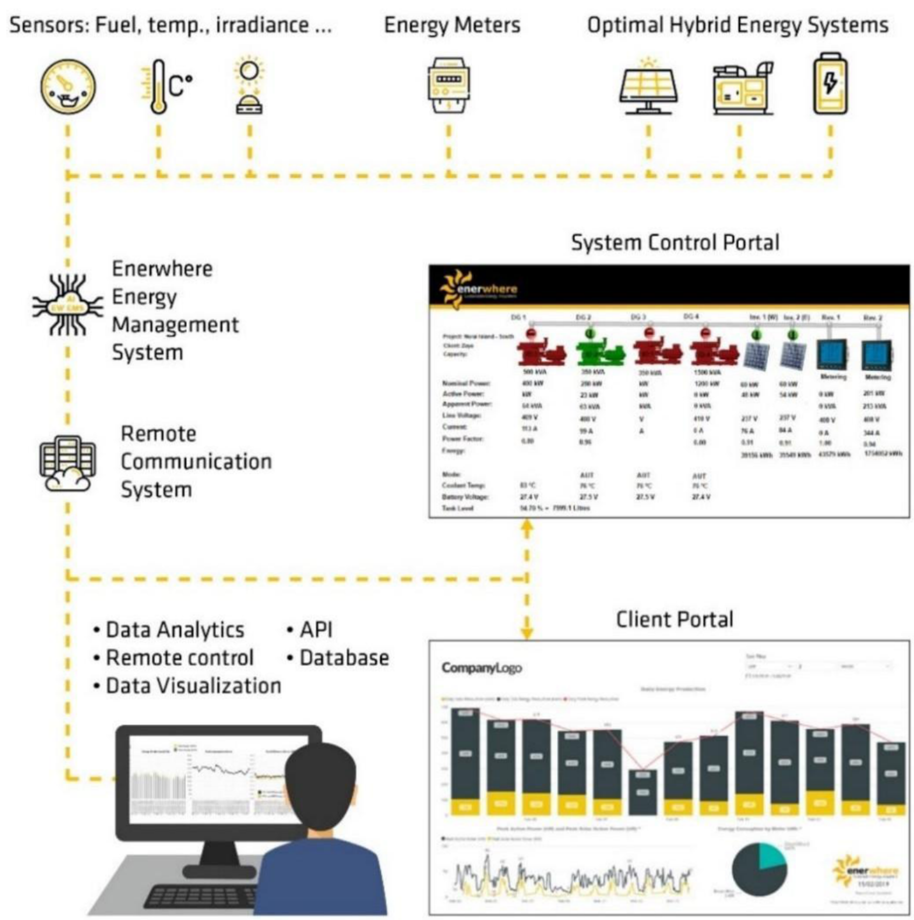


Figure 2—Online Monitoring System

The layout below shows the mobile solar system's components. Note that the actual system installation does not include a trailer, as the rig moves infrequently (on average, once every two months). A trailer-based solution would be more suitable for applications that relocate more frequently, such as once or twice per week, facilitating relocation by eliminating the need for cranes and lifting equipment.

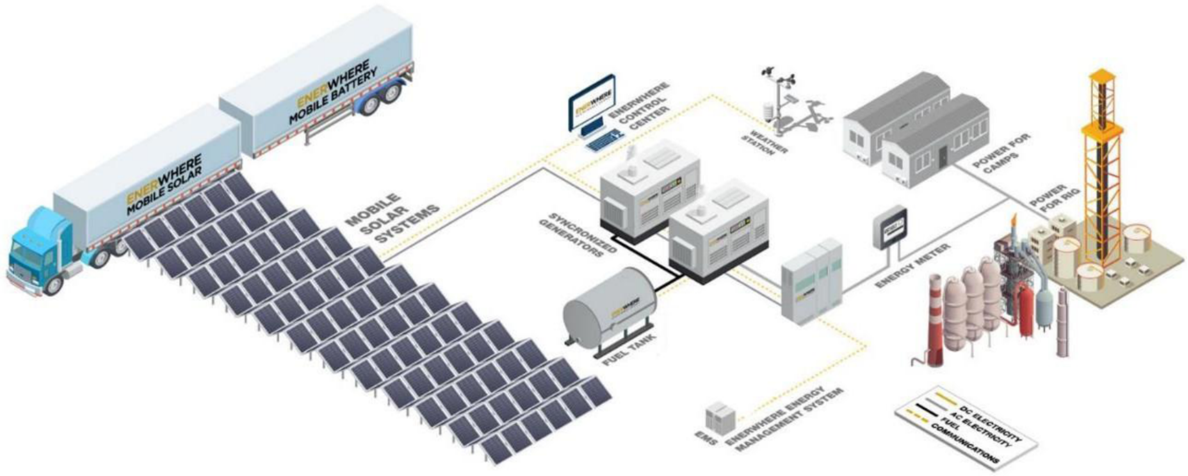


Figure 3—System Overall Layout

## Remote Monitoring via Online Platform

The online platform developed by the Provider enabled remote control and real-time monitoring of the mobile solar hybrid system. During the design phase of the project, it supported high-resolution data collection, allowing for detailed simulations and analysis of load demands. This capability was instrumental in establishing an accurate energy baseline for the drilling rig. The energy baseline allowed the Provider to establish the peak loads of each operation taking place on the drilling rig.

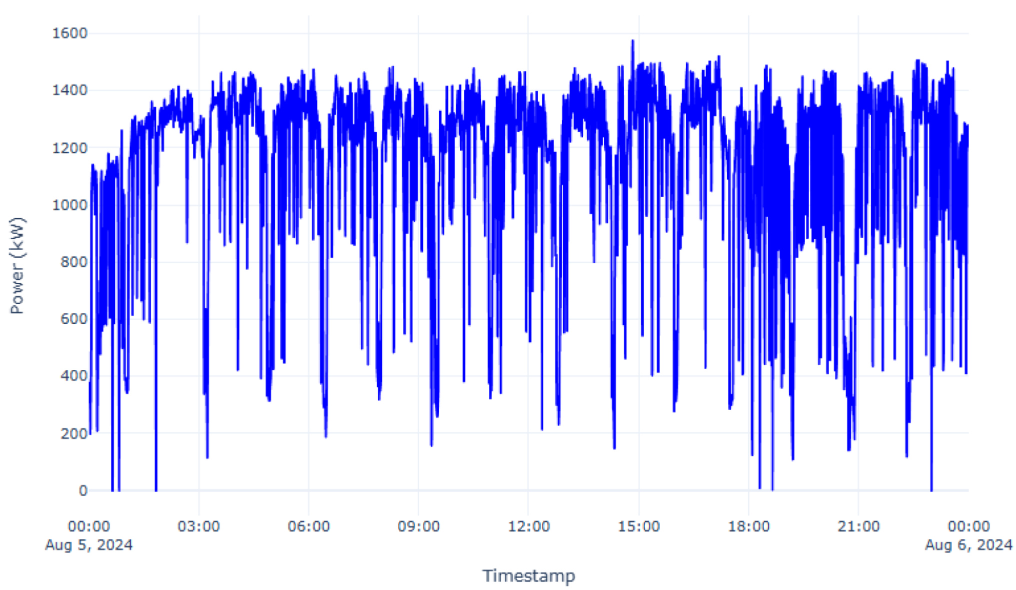


Figure 4—Rig Power Consumption at Drilling

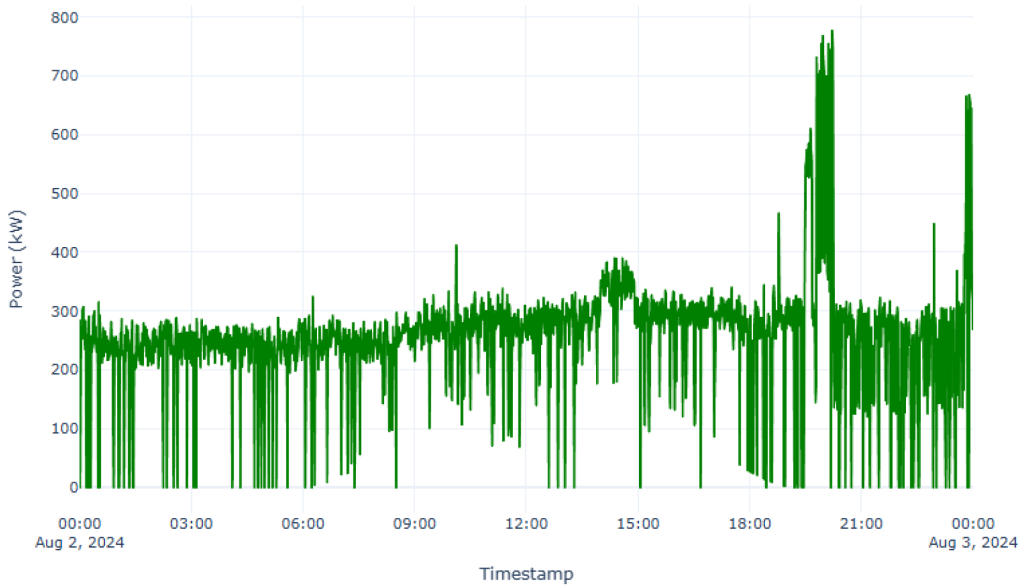


**Figure 5—Rig Power Consumption no Drilling**

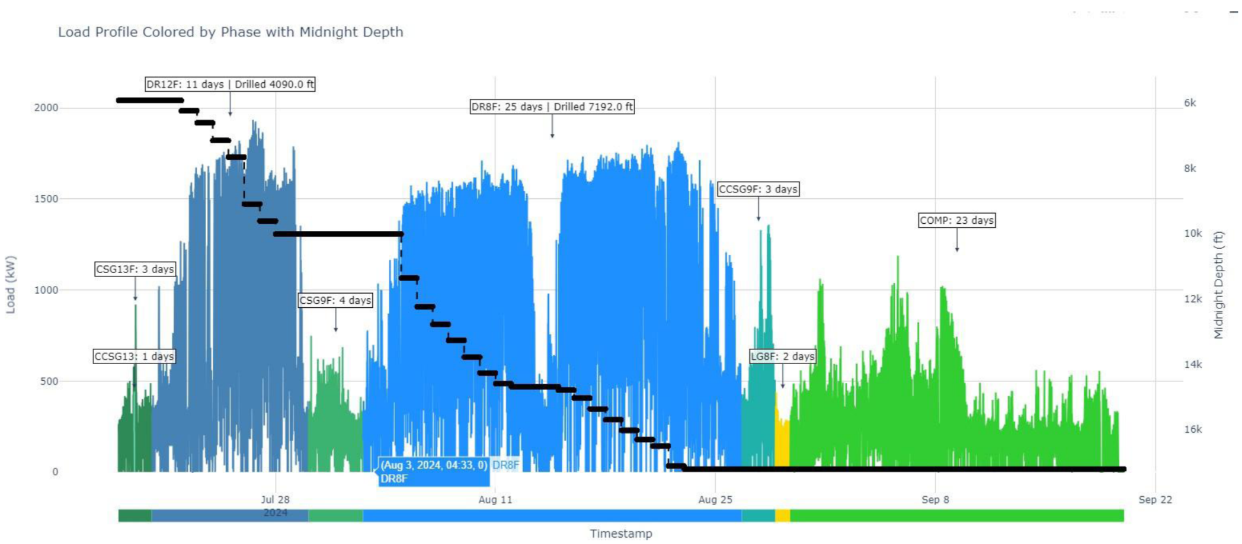


**Figure 6—Rig Power Consumption at Different Stages**

During the operation phase, the remote monitoring and control system continuously tracked the operational status and performance of both the diesel generators and the solar PV array. This level of digitization in energy generation and management marked a significant advancement toward decarbonization. With real-time insights into energy usage and system performance, the Operator could accurately evaluate initial emission levels and overall efficiency. Moreover, the constant stream of performance data enabled ongoing analysis, helping to identify areas for further optimization. By fine-tuning the interaction between energy supply and consumption patterns, the system could achieve greater efficiency and a progressively lower carbon footprint.

One of the major limitations of conventional off-grid systems is their rigid design. Once deployed, these systems offer limited adaptability or visibility beyond basic refueling and scheduled maintenance. This inflexibility becomes a significant issue when dealing with various fluctuations in energy demand. Since power requirements vary throughout the operation of the drilling rig, a system capable of adapting dynamically to these changes is essential. By incorporating flexible, data-driven energy management, off-grid systems can be continuously optimized to deliver reliable, efficient performance under varying load conditions.


Figure 7—Monitoring and Control Architecture

## Deployment and Operational Procedure

### Deployment Methodology

The system was installed and commissioned at an off-grid drilling rig location in Abu Dhabi in July 2024. Site preparation included surface leveling and access routing. Solar containers were deployed adjacent to the rig and connected to the rig's Low Voltage (LV) panel through the hybrid controller. The setup ensured full operational continuity with no interference to rig start-up or ongoing activities.

The Energy Management System (EMS) was configured based on site-specific load profiles, with remote fine-tuning capability via the Provider's digital platform.

### System Operation

The EMS continuously monitored rig loads and solar generation to determine the optimal power source mix. Solar was prioritized during daylight hours, with diesel generators running intermittently or continuously depending on load demand and solar output.

The system automatically curtailed solar input when necessary to maintain grid stability and prevent underloading of diesel generators. Generators operated at their efficiency sweet spot, reducing fuel consumption and mechanical stress.

The system's key performance indicators included:

- Solar share (% of total energy)
- Diesel generator runtime and loading factor
- Real-time system availability and fault detection
- Remote accessibility and control reliability

## Results and Performance Evaluation

### Energy Contribution and Fuel Savings

During the 5-month pilot:

- Total rig energy consumption: **1,630 MWh**
- Solar energy contribution: **117 MWh (7%)**
- Diesel offset. & $CO_2$ emission reduction approx. **7%**
- **5 to 8%** of the total energy consumption was generated from solar energy during high load demand (drilling phases), while the percentage increased up to **30%** during low load demand phases

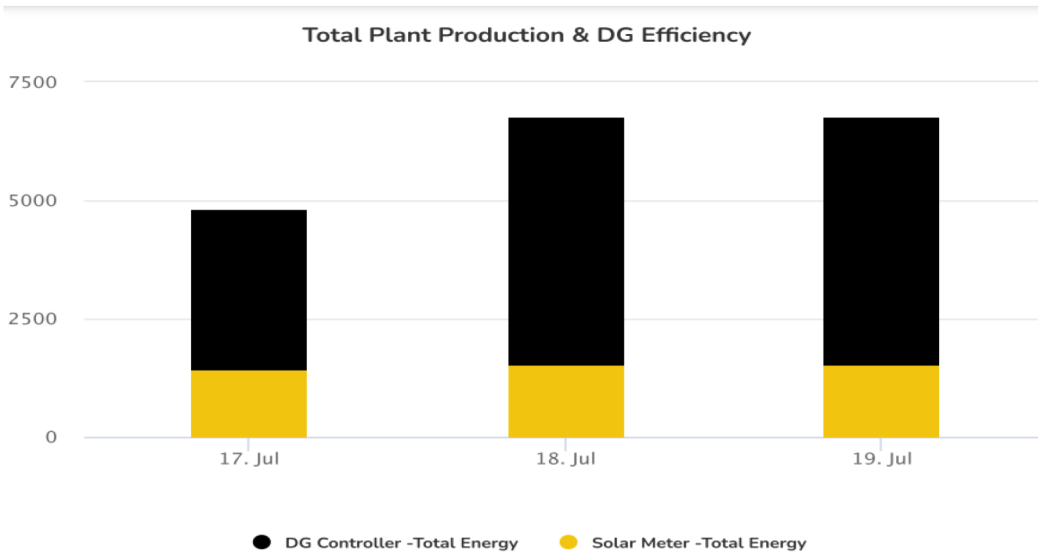


**Figure 8—Actual Rig Energy Consumption (Monthly)**

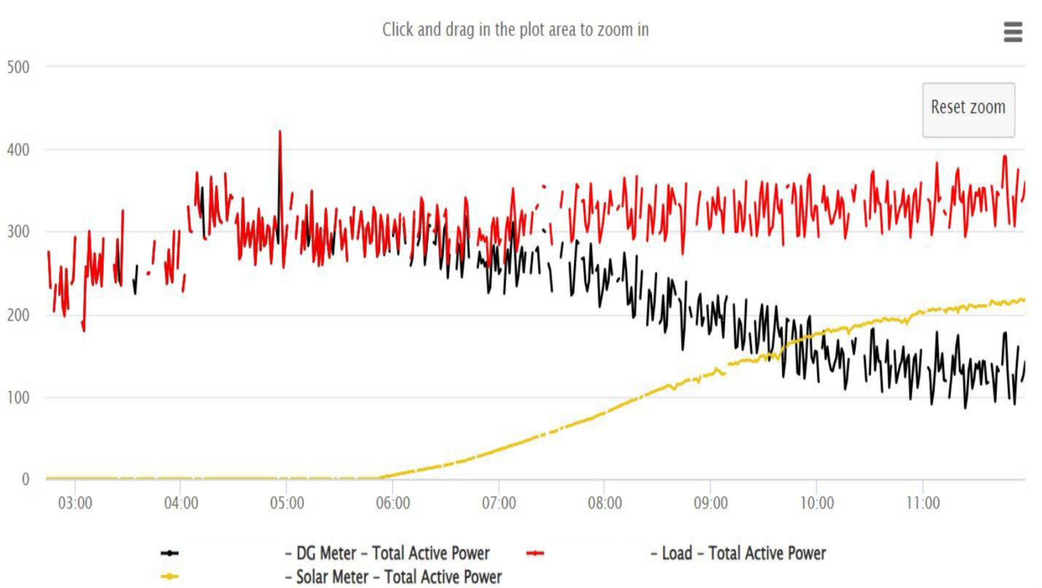


**Figure 9—Rig High Resolution Load Profile**

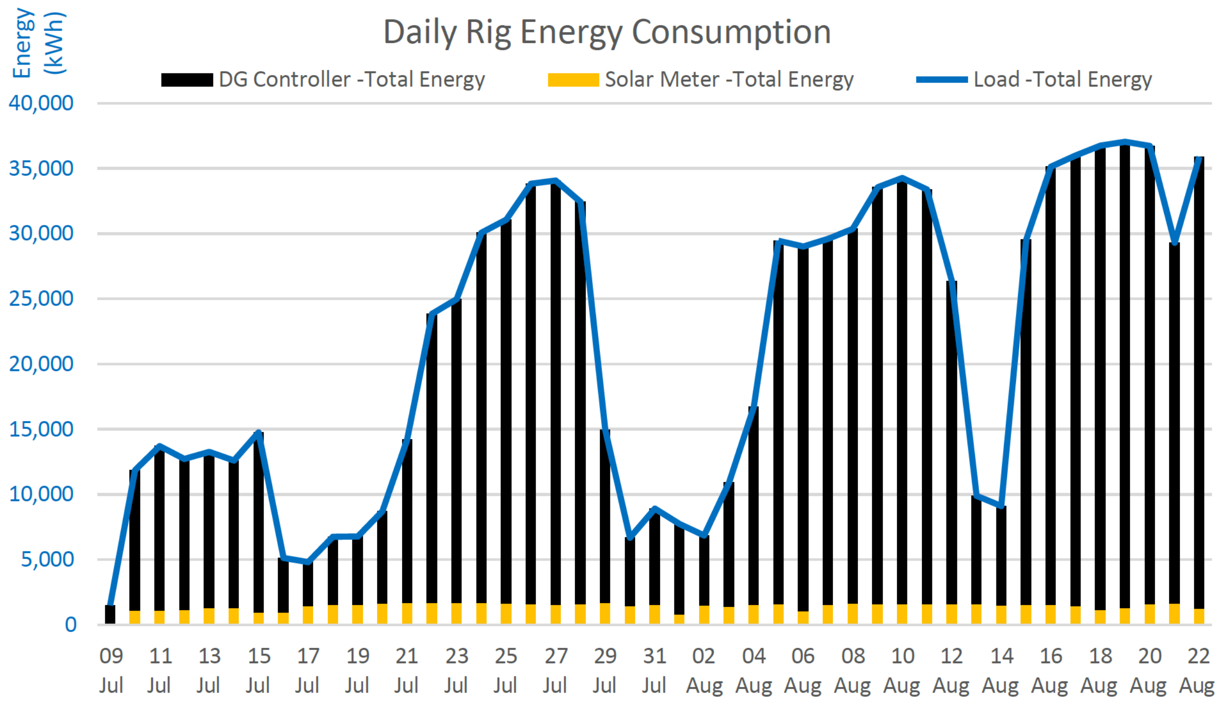


**Figure 10—Daily Energy Profile of the Rig**

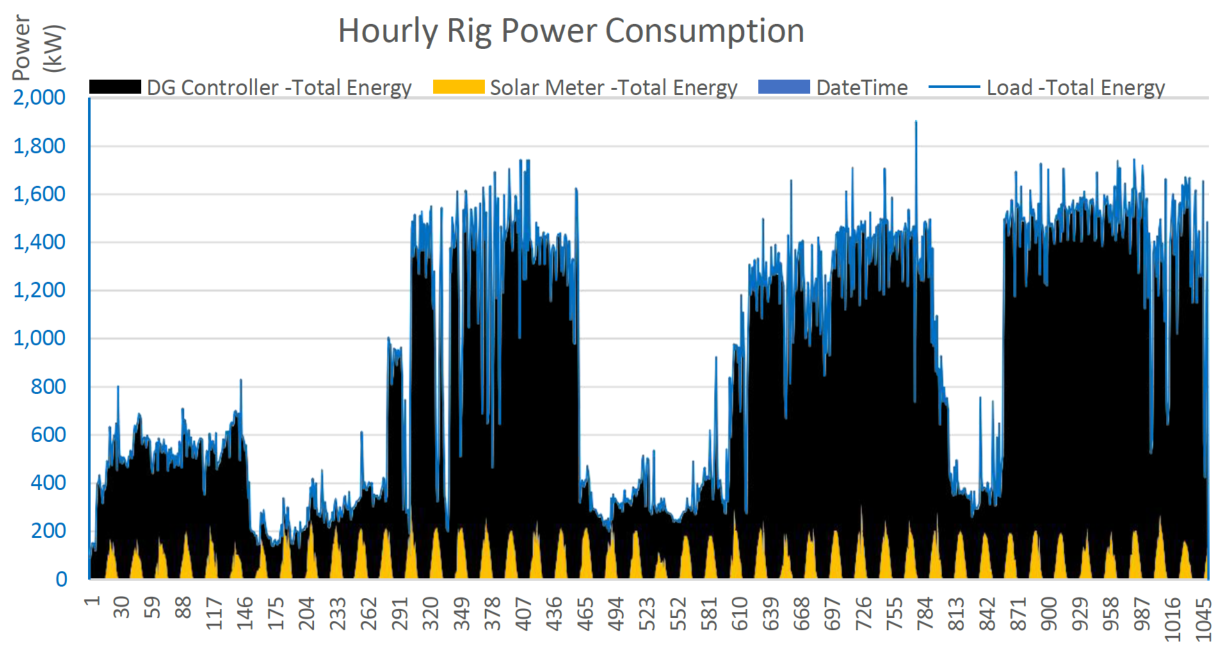


**Figure 11—Rig Hourly Load Profile**

## Operational Benefits

## Key benefits included:

1. **Improved Fuel Efficiency**
   Diesel generators operated more efficiently, reducing total fuel use and wear.
2. **Reduced Generator Runtime & Maintenance**
   Generator maintenance intervals increased due to fewer operating hours.
3. **Lower Acoustic Impact**
   Reduced generator usage significantly lowered onsite noise levels.

4. **Remote Monitoring & Diagnostics**
   Enabled proactive troubleshooting, reducing need for field visits.
5. **Rapid Mobilization & Scalability**
   Mobilization time under 6 hours; easily relocatable between rigs.
6. **Safe, Reliable Power Supply**
   Zero HSE incidents; maintained continuous power across operational phases.

## Discussion and Lessons Learned

The pilot demonstrated the hybrid system's adaptability and effectiveness under real-world rig operating conditions. Several lessons emerged:

- **Load Profile Complexity:** Drilling rigs exhibit sharp, high-load demands; thus, even small solar contributions require careful coordination.
- **System Optimization Needs:** Future deployments should integrate BESS to further enhance solar contribution and reduce diesel reliance.
- **Environmental Challenges:** High ambient temperatures and dust accumulation necessitate frequent solar panel cleaning and robust component selection.
- **Scalability Potential:** With minor design tweaks, the system can be replicated fleet-wide, increasing overall decarbonization impact.

## Background Literature Review

The integration of renewable energy solutions into oil and gas operations has been the subject of increasing research over th e past decade, driven by both environmental policy pressures and operational cost considerations. Hybrid energy systems, combining solar photovoltaic (PV) generation with conventional diesel generation, have been successfully demonstrated in various remote and off-grid applications, including mining operations, military camps, and isolated communities. Studies by the International Renewable Energy Agency (IRENA) and other industry bodies have highlighted the potential for hybrid systems to reduce fuel consumption by 10-40%, depending on site-specific load profiles and solar resource availability.

Within the oil and gas sector, applications remain relatively limited due to concerns over reliability, high and variable power demands, and the logistical complexities of integrating solar systems into mobile operations. However, technological advancements in energy management systems (EMS), modular PV container design, and the availability of high-efficiency panels are rapidly improving feasibility. Several pilot projects in regions with abundant solar resources—such as the Middle East, Australia, and West Africa—have shown that even a modest share of solar penetration can lead to measurable reductions in fuel usage, maintenance frequency, and $CO_2$ emissions.

This project builds upon that body of work by deploying a containerized, ultra-mobile solar-diesel hybrid system on an active land drilling rig in Abu Dhabi, UAE. Unlike most previous trials, which have focused on semi-permanent facilities, this deployment addressed the unique challenge of high mobility, extreme climate conditions, and the need for minimal disruption to critical drilling operations. The findings presented here not only confirm the operational viability of such systems but also demonstrate the potential for scalability across multiple rigs within the region.[2]

## Conclusion and Future Outlook

Beyond the immediate operational benefits, the implications of this project extend to policy, investment, and technology development within the upstream energy sector. A successful transition to hybrid systems can enhance energy security, reduce environmental impact, and create opportunities for local manufacturing and employment. Furthermore, lessons learned from this pilot can inform the development of technical standards and operational guidelines for hybrid system integration in other challenging environments worldwide. The successful operation of a mobile solar-diesel hybrid system at an active drilling rig marks a regional first and a major step toward sustainable upstream practices. Though solar provided only 7% of the rig's energy, the system validated its potential for improving operational efficiency, reducing emissions, and aligning with long-term net-zero targets.

**Key takeaways:**

- Demonstrated diesel savings and $CO_2$ reductions in a high demand setting
- Maintained continuous rig power with no operational disruption
- Enabled data-driven decision-making via remote monitoring

**Future improvements include:**

- Improvement of solar potential per mobile container
- Increase in number of mobile containers deployed
- Integration of BESS for greater solar penetration
- Enhancement of skid-mounted system to further improve deployment time of mobile containers
- Wider deployment across multiple rigs
- Potential adaptation for powering auxiliary loads (e.g., mud pumps, electrical submersible pumps, mobile camps, base camps)
- Manufacturing and assembly in the United Arab Emirates to further support local content

This project exemplifies how modular, mobile hybrid energy systems can transform upstream energy practices—achieving both operational excellence and environmental stewardship.

The deployment of a mobile solar hybrid system on a live drilling rig marks a significant step forward in decarbonizing off-grid operations within the oil and gas industry. This innovative approach delivers multiple benefits, including improved energy efficiency, enhanced reliability, reduced carbon emissions, lower operating costs, and strengthened HSE performance—while directly supporting broader sustainability targets.

Over a six-month period, the pilot project achieved more than 7% diesel reduction. These results demonstrate the system's strong potential for replication across other off-grid drilling operations. Future implementations that incorporate higher solar penetration and Battery Energy Storage Systems (BESS) could enable emissions reductions exceeding 20% per site.

Another key insight relates to data-driven decision-making. The remote monitoring platform not only provided operational oversight but also generated a valuable dataset that can be used to model future deployments, optimize equipment sizing, and anticipate maintenance needs.

The project further demonstrated the importance of designing for maintainability and ease of relocation. The modular, plug -and-play nature of the solar containers meant that site moves could be accomplished in under six hours, a critical factor for rigs that regularly relocate between well sites. This mobility significantly reduces the downtime typically associated with generator-only setups and offers a competitive advantage in terms of operational flexibility.

Finally, the pilot underlined that even modest solar penetration rates can deliver meaningful environmental benefits in high-load operations. A 7% solar share, while seemingly small, translated into substantial diesel savings and avoided emissions—benefits that will compound when replicated across a larger fleet of rigs. This reinforces the role of hybrid systems as an incremental yet impactful pathway toward decarbonization in hard-to-abate sectors.

Looking ahead, this mobile hybrid power solution can be expanded beyond drilling rigs to support other remote oilfield applications— such as base camps, mobile camps, and electrically powered artificial lift systems —delivering scalable and impactful progress toward net-zero goals.